

SMART PARKING PLATFORM

Project Management Plan
WE ARE \$oftware ¢orp

Version 0.4 - Homework 1-4

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1. Vision and Scope

WE ARE \$oftware ¢orp. is developing the Smart Parking Platform as an internal commercial product. The web and mobile platform will help drivers locate available parking, reserve spaces, pay digitally, and navigate to participating garages. Parking operators will manage facilities, pricing, availability, reservations, and reporting.

ID	Requirement
FR-01	Register and authenticate users; support password reset and logout.
FR-02	Search for parking by location, destination, date, or time.

FR-03	Display parking locations on an interactive map.
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1.1 Scope

In Scope	Out of Scope
Registration/authentication; real-time availability; map/search; facility details; reservations; payments; history/receipts; operator garage/pricing/inventory management; reports; notifications; mapping/navigation integration.	Physical parking hardware; garage construction/operation; parking enforcement/towing; proprietary GPS infrastructure; physical equipment maintenance.

1.2 Stakeholders and Constraints

Stakeholders	Key Constraints
Drivers; parking operators; city transportation departments; system administrators; finance/billing personnel; mobile users; third-party payment providers.	Fixed semester timeline; third-party API/service dependencies; security/privacy requirements; city ordinances and laws; unlimited theoretical development budget.

2. Competitive Analysis

Platform	Strengths / Existing Capabilities	Project Differentiation Opportunity
SpotHero	Parking search, reservations, prepaid parking, and operator tools.	Create a consistent workflow connecting live availability, reservations, payments, navigation, and operator updates.
ParkWhiz	Parking reservations, payments, and operator reporting/integrations.	Emphasize standardized facility data and a consistent experience across participating garages.
ParkMobile	Reservations plus parking-management, pricing, analytics, and provider tools.	Connect driver and operator workflows through the same real-time parking data.

3. Software Requirements Specification

3.1 Functional Requirements

FR-04	Display facility details, pricing, restrictions, and availability.
FR-05	Display real-time parking availability from supported data sources.
FR-06	Create and cancel parking reservations.
FR-07	Process digital payments through an approved payment provider.
FR-08	Display reservation history, confirmations, and receipts.
FR-09	Send reservation and account notifications.
FR-10	Launch supported mapping/navigation for a selected facility.
FR-11	Allow operators to add, edit, and deactivate facilities.
FR-12	Allow operators to manage pricing, inventory, and availability.
FR-13	Allow operators to review reservations and facility activity.
FR-14	Provide occupancy and utilization reporting.
FR-15	Provide authorized financial and transaction reporting/export.

3.2 Non-Functional Requirements

ID	Requirement
NFR-01	Encrypted communications and secure password storage.
NFR-02	Role-based access control.
NFR-03	Target 99.9% availability, excluding planned maintenance and thirdparty outages.
NFR-04	Common requests should normally respond within 3 seconds, excluding third-party delays.
NFR-05	Responsive web/mobile interface.
NFR-06	Scalable architecture for growth in users, facilities, and transactions.
NFR-07	Audit logging for important administrative and financial actions.
NFR-08	Graceful handling of third-party service failures.

4. Use Cases

Use Case	Actor	Main Flow
UC-01 Register Account	Driver / Operator	Enter account information -> validate -> create account.
UC-02 Log In	Driver / Operator / Admin	Enter credentials -> validate -> open authorized dashboard.
UC-03 Reset Password	Registered User	Request reset -> verify identity -> create new password.
UC-04 Search Parking	Driver	Enter destination/date/time -> display matching facilities.
UC-05 View Parking Map	Driver	Open map -> display matching facilities -> select facility.
UC-06 View Facility Details	Driver	Select facility -> display rates, restrictions, details, and availability.
UC-07 Create Reservation	Driver	Select facility/time -> validate inventory -> create reservation.
UC-08 Pay for Reservation	Driver	Choose payment -> authorize transaction -> record payment and receipt.
UC-09 Cancel Reservation	Driver	Select eligible reservation -> confirm cancellation -> update inventory/status.
UC-10 View History / Receipt	Driver	Open history -> select reservation/transaction -> display details.
UC-11 Navigate to Facility	Driver	Select directions -> send destination to mapping/navigation service.
UC-12 Add Parking Facility	Operator	Enter facility details -> validate -> save facility.
UC-13 Edit Facility / Pricing	Operator	Edit facility/rates/restrictions -> validate -> save changes.
UC-14 Manage Availability	Operator	Update parking inventory -> validate -> refresh availability.
UC-15 Review Reports	Operator / Finance	Choose report/date range -> aggregate data -> display/export report.

5. Work Breakdown Structure

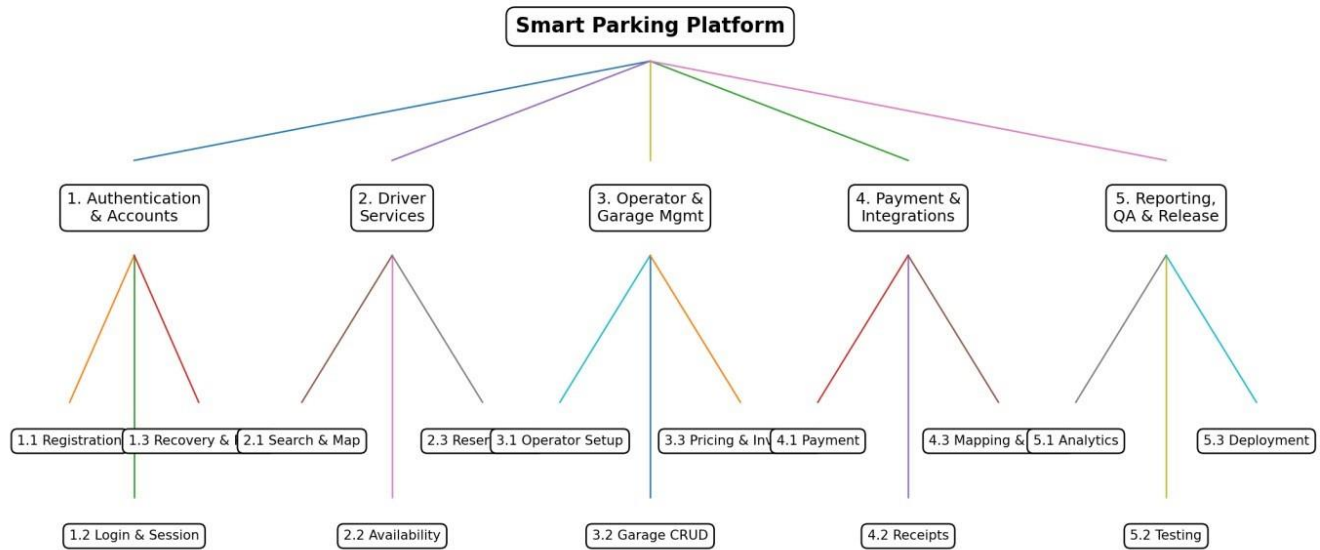


Figure 1. Smart Parking Platform Work Breakdown Structure (three levels or more).

6. Project Schedule

Deliverable	Planned Dates
Requirements Baseline	Sep 1-11
Architecture & UI Design	Sep 8-18
Authentication & Accounts	Sep 14-25
Operator / Garage Management	Sep 21-Oct 9
Driver Search, Map & Availability	Sep 28-Oct 16
Reservations & Payments	Oct 12-30
Reporting & Notifications	Oct 26-Nov 6
System Integration	Nov 2-13
QA & Security Testing	Nov 9-27
Final Release & Documentation	Nov 23-Dec 4

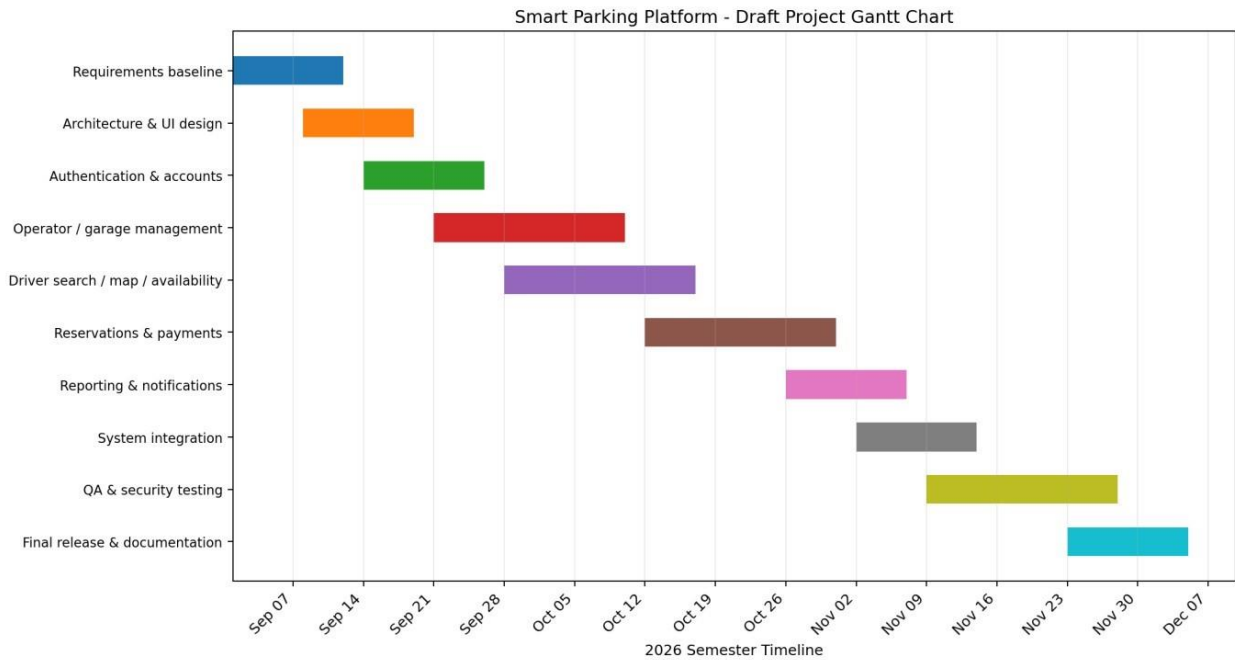


Figure 2. Draft semester Gantt chart.

7. Product Backlog and Sprint 1

Product backlog: 45 items total - 5 Login, 15 UI, 15 Backend, and 10 Reporting.

Category	Backlog Items
Login	LOG-01 Driver Login; LOG-02 Operator Login; LOG-03 User Registration; LOG-04 Password Reset; LOG-05 Session Logout
UI	UI-01 Driver Dashboard; UI-02 Operator Dashboard; UI-03 Parking Search Interface; UI-04 Interactive Parking Map; UI-05 Parking Facility Details; UI-06 Real-Time Availability Display; UI-07 Reservation Screen; UI-08 Payment Checkout Screen; UI-09 Reservation Confirmation Screen; UI-10 Reservation History Screen; UI-11 Receipt View; UI-12 Driver Profile Settings; UI-13 Garage Management Interface; UI-14 Pricing Management Interface; UI-15 Operator Reporting Dashboard
Backend	BE-01 Driver Account Creation API; BE-02 Operator Account Creation API; BE-03 Authentication API; BE-04 Password Recovery Process; BE-05 Parking Facility Search API; BE-06 Real-Time Availability Service; BE-07 Garage Creation Process; BE-08 Garage Update Process; BE-09 Parking Inventory Management; BE-10 Reservation Creation Process; BE-11 Reservation Cancellation Process; BE-12 Payment Processing Integration; BE-13 Receipt Generation Process; BE-14 Mapping & Navigation Integration; BE-15 Notification Service
Reporting	REP-01 Current Occupancy Report; REP-02 Daily Occupancy Report; REP-03 Weekly Occupancy Report; REP-04 Reservation Activity Report; REP-05 Revenue Summary Report; REP-06 Transaction History Report; REP-07 Garage Utilization Report; REP-08 Peak Parking Hours Report; REP-09 Driver Reservation History Report; REP-10 Financial Report Export

7.1 Sprint 1 Backlog

ID	Sprint 1 Item
LOG-02	Operator Login
LOG-03	User Registration
BE-02	Operator Account Creation API
BE-03	Authentication API
BE-06	Real-Time Availability Service
BE-07	Garage Creation Process
BE-08	Garage Update Process
BE-09	Parking Inventory Management

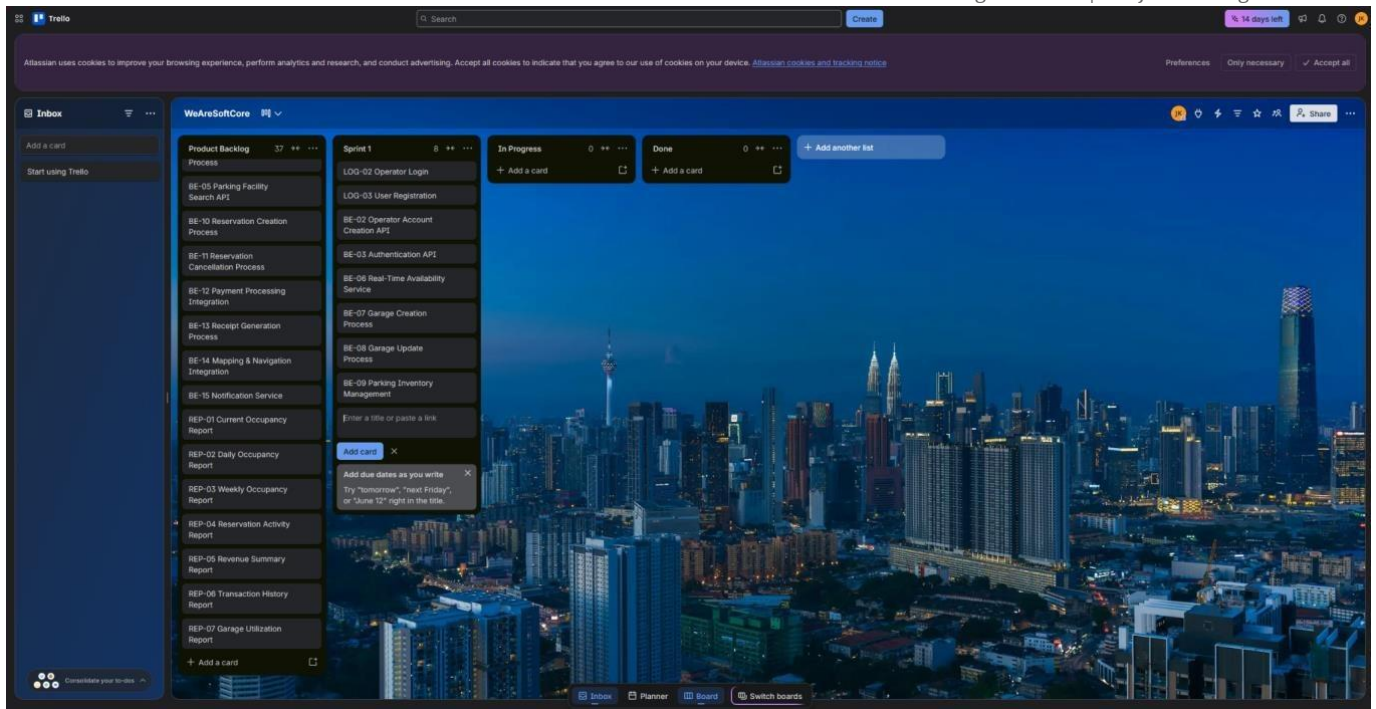


Figure 3. Trello board showing the Product Backlog and Sprint 1.

8. Risk Management

Risk management identifies uncertain events that could affect project objectives, assesses their probability and impact, and defines an appropriate response. The Smart Parking Platform risks below are grouped into the four categories used for this project: technical, schedule, financial, and people.

8.1 Technical Risks

- T-01 - Third-party mapping, payment, or availability APIs become unavailable or change unexpectedly.
- T-02 - Real-time parking availability data is delayed, inaccurate, or inconsistent across participating garages.
- T-03 - Authentication, payment, or personal-data vulnerabilities expose sensitive user information.
- T-04 - System integration between mobile/web interfaces, backend services, and external providers fails or performs poorly.

8.2 Schedule Risks

- S-01 - Third-party integration work takes longer than planned and delays reservations, payments, or navigation features.
- S-02 - Authentication or operator/garage management work slips and delays dependent driver-facing functionality.
- S-03 - Defects discovered during QA and security testing require significant rework near the end of the semester.
- S-04 - Scope growth or late requirement changes reduce time available for integration, documentation, and final release.

8.3 Financial Risks

- F-01 - Payment, mapping, cloud, messaging, or other service fees exceed the project cost assumptions.
- F-02 - Unexpected rework increases development and testing effort beyond the planned budget.
- F-03 - Scaling infrastructure for users, facilities, and transactions creates higher recurring hosting costs.
- F-04 - Security, privacy, or city-compliance changes require additional tools, services, or implementation work.

8.4 People Risks

- P-01 - A key team member becomes unavailable during a critical development or testing period.
- P-02 - Skill gaps in APIs, security, payments, or mobile/web integration slow implementation or increase defects.
- P-03 - Miscommunication between development, project management, and stakeholders causes conflicting assumptions or rework.
- P-04 - Uneven workload or unclear ownership creates bottlenecks and lowers team productivity.

9. Risk Register

ID	Category	Risk Description	Prob.	Impact	Priority	Owner	Response	Mitigation / Contingency	Status
R-01	Technical	Third-party API outage/change	3	4	High	Integration Lead	Mitigate	Use abstraction layers, health checks, retry/fallback behavior, and provider monitoring.	Open
R-02	Technical	Inaccurate real-time availability	3	4	High	Backend Lead	Mitigate	Validate timestamps/data, define refresh rules, flag stale feeds, and reconcile operator updates.	Open
R-03	Technical	Security/privacy vulnerability	2	5	High	Security Lead	Mitigate	Apply secure authentication, encryption, RBAC, code review, security testing, and audit logging.	Open
R-04	Schedule	Third-party integration delay	3	4	High	Project Manager	Mitigate	Start integrations early, use mocks/sandboxes, track dependencies weekly, and keep contingency time.	Open
R-05	Schedule	Late QA defects cause rework	3	4	High	QA Lead	Mitigate	Shift testing earlier, automate regression tests, and reserve time for defect correction.	Open
R-06	Schedule	Scope growth / late changes	3	3	Medium	Product Owner	Avoid	Require change review, protect the agreed scope, and defer nonessential additions.	Open
R-07	Financial	Third-party/service costs rise	2	3	Medium	Finance Lead	Mitigate	Track usage and cost assumptions; set alerts and evaluate lower-cost tiers or alternatives.	Open
R-08	Financial	Rework increases labor cost	3	3	Medium	Project Manager	Mitigate	Use acceptance criteria, peer reviews, early demos, and defect tracking to reduce avoidable rework.	Open

R-09	People	Key team member unavailable	2	4	Medium	Project Manager	Mitigate	Cross-train team members, document work, share repository access, and identify backup owners.	Open
R-10	People	Miscommunication causes rework	3	3	Medium	Project Manager	Mitigate	Use documented decisions, Trello status updates, meeting notes, and clear escalation paths.	Open

10. Communication Plan

Audience / Team	Cadence	Participants	Method	Primary Content	Owner
Development Team	Daily stand-up	Developers, Technical Lead	10-15 minute stand-up / team chat	Yesterday/today, blockers, dependency changes, immediate technical risks	Technical Lead
Project / Sprint Team	Weekly progress meeting	Project Manager, Product Owner, team leads	Meeting + Trello review	Schedule status, Sprint progress, open risks/issues, decisions, next-week priorities	Project Manager
QA & Security	Weekly during development; 2x weekly during QA phase	QA Lead, Security Lead, developers	Defect dashboard + test review	Test results, defect trends, security findings, release blockers	QA Lead
Integration Team	Weekly; ad hoc for provider changes	Backend/Integration Lead, relevant developers	Integration checklist + API status notes	External service status, interface changes, failures, fallback actions	Integration Lead
Stakeholders / Sponsors	Biweekly and at major milestones	Project Manager, Product Owner, key stakeholders	Status report + demo when applicable	Milestones, schedule health, major risks, completed functionality, decisions needed	Project Manager
Finance / Reporting	Monthly and before release	Finance Lead, Project Manager, Product Owner	Cost/usage summary + report review	Service-cost assumptions, transaction/reporting readiness, financial risks	Finance Lead
Release Readiness	Before final release / milestone gate	Project Manager, Technical Lead, QA/Security Leads, Product Owner	Go/no-go checklist + release report	Open critical defects, security status, documentation, rollback/contingency plan, approval	Project Manager

11. Roles & Resource Allocation

Role	Fictional name	Skill	Allocation	Hours	Rate/hr	Cost (\$)
Project Manager	Alex Morgan	Senior	35%	168	\$60	10,080
Product Owner	Taylor Reed	Senior	20%	96	\$55	5,280
Technical / Backend Lead	Jordan Lee	Senior	75%	360	\$65	23,400
Frontend / UX Developer	Casey Nguyen	Mid-level	75%	360	\$50	18,000
Integration / DevOps Lead	Morgan Patel	Senior	50%	240	\$65	15,600
QA / Security Lead	Riley Chen	Mid-level	50%	240	\$50	12,000
Finance / Reporting Lead	Sam Rivera	Mid-level	20%	96	\$45	4,320

12. Resource & Cost Plan

12.1 Bottom-Up Cost Estimate

Category	Calculation / assumption	Cost (\$)
Labor	Staffing from Section 11	88,680
Cloud infrastructure	4 months x \$350	1,400
Mapping / payment / notification APIs	4 months x \$200	800
Licenses and developer tools	One-time allowance	600
Security / testing services	One-time allowance	800
Subtotal before contingency		92,280
Contingency reserve (10%)	10% of subtotal	9,228
Total planned budget		101,508

12.2 Budgeted Monthly Cashflow

Month	Labor (\$)	Other (\$)	Reserve (\$)	Total (\$)
September	17,736	1,150	1,846	20,732
October	31,038	550	3,230	34,818
November	26,604	1,350	2,768	30,722
December	13,302	550	1,384	15,236
TOTAL	88,680	3,600	9,228	101,508

13. RACI Matrix

Activity	PM	PO	TL	UX	INT	QA	FIN
Requirements / scope	A	R	C	C	I	I	I
Architecture / backend	C	I	A/R	C	C	C	I
Driver UI / search / map	I	C	C	A/R	C	C	I
Garage / availability services	I	C	A/R	C	R	C	I
Reservations / payments	I	C	A	R	R	C	C
System integration / deployment	C	I	R	C	A/R	C	I
QA and security testing	I	C	C	C	C	A/R	I
Reporting / financial export	I	C	R	C	C	C	A
Budget tracking / change control	A	C	C	I	C	I	R
Final release approval	R	A	C	I	C	C	I

R = Responsible A = Accountable C = Consulted I = Informed

PM = Project Manager PO = Product Owner TL = Technical Lead UX = Frontend/UX INT = Integration/DevOps QA = QA/Security FIN = Finance/Reporting